

Customer Churn & Retention Analytics

Business Problem Statement :

Customer churn is a major challenge for subscription-based businesses because losing existing customers can reduce recurring revenue and increase customer acquisition costs.

The company has customers with different segments, subscription plans, regions, service-usage levels, payment methods and satisfaction levels. The business needs to understand which customer groups are more likely to churn, why customers are leaving, and which areas are creating the highest financial risk.

The challenge is therefore not only to measure the overall churn rate, but also to identify churn patterns, high-risk customer segments and revenue at risk.

This project analysis customer and subscription data to understand churn across different business dimensions, including customer segment, subscription plan, region, tenure, service usage, payment method, customer satisfaction and churn reason.

The analysis also evaluates revenue generated by different subscription plans and estimates the monthly revenue at risk from churned customers. SQL analysis was used to answer business questions and identify patterns, while Excel and Power BI were used for analysis, KPI reporting and interactive visualization.

Objectives :

Identify the major churn patterns and high-risk customer groups, quantify the revenue exposed to churn, and provide data-driven recommendations that can help the business improve customer retention and protect recurring revenue.

Tools :

Excel was used for data cleaning, validation, PivotTables and exploratory analysis. MySQL was used to perform business analysis using joins, aggregations, CTEs and window functions. Power Query was used for data transformation, while Power BI and DAX were used to build the data model, calculate KPIs and develop the interactive customer churn dashboard.

Dataset Overview :

The Customer Churn & Retention Analytics project uses a relational customer dataset designed to represent a subscription-based business. The dataset contains customer information along with subscription, transaction, service usage, support, feedback, churn and marketing campaign data.

The data is organized into 7 main analytical tables, allowing customer behaviour to be analyze from multiple business perspectives.

Note : The dataset was created as a realistic sample business dataset for portfolio analysis. The larger dataset was designed to represent a more realistic data volume, while a smaller representative dataset was used for MySQL query execution because of local MySQL Workbench import limitations.

Excel Data Cleaning & Exploratory Analysis

Data Cleaning :

The raw customer churn dataset contained data-quality issues that could affect the accuracy of the analysis. Excel was used as the initial data preparation and validation tool before performing further analysis in MySQL and Power BI.

The main objective of data cleaning was to ensure that the data was accurate, consistent, complete and suitable for analysis.

Data Cleaning Activities :

Data Quality Issue	Cleaning Approach
Duplicate records	Identified and removed duplicate records where applicable
Missing values	Identified blank/null values and handled them appropriately
Missing Sales Person / Payment Method	Reviewed missing values and handled them without making unsupported assumptions
Inconsistent city names	Standardized inconsistent city/category values
Invalid age values	Identified unrealistic age values and handled them according to business rules
Data types	Checked and corrected fields such as dates and numeric values
Category consistency	Reviewed categorical fields for inconsistent values

KPI Analysis :

After cleaning the data, Excel was used to calculate the key performance indicators required to understand customer retention and churn.

- Total Customers
- Active Customers
- Churned Customers
- Churn Rate
- Retention Rate
- Monthly Revenue at Risk
- Average Monthly revenue per customer
- Average Customer Tenure
- Failed Payment Rate
- Average Customer Satisfaction

PivotTable Analysis :

PivotTables were created after cleaning the data to perform exploratory analysis and identify patterns in customer churn.

Rather than creating PivotTables randomly, they were designed around the relationships between customer attributes, subscription information, customer behaviour and churn outcomes.

- Churn by Region
- Churn by Customer Segment
- Churn by Subscription Plan
- Churn by Contract Type
- Churn by Tenure Group
- Churn by Service Usage Level
- Churn by Payment Method
- Churn by Customer Satisfaction
- Churn by Customer Sentiment
- Churn by Churn Reason
- Revenue at Risk by Segment
- Monthly Churn Trend
- Campaign Performance

Power BI Analysis & Dashboard Development

Data Model :



The Power BI data model is shown in the diagram above connects the customer, subscription, transaction, service usage, support, feedback and churn data using their respective keys.

The model allows the dashboard to move from a high-level business KPI to detailed customer and subscription analysis without treating each dataset independently.

Dashboard KPI'S :

The main dashboard provides an executive-level view of customer retention performance.

- Total Customers
- Churned Customers
- Churn Rate
- Monthly Revenue at Risk
- Retention Rate

The dashboard currently shows approximately 10000 customers, 2875 churned customers, 28.75% churn rate and ₹4.62M monthly revenue at risk.

Interactive Dashboard :



Power BI was used as the final reporting and visualization layer of the project. The cleaned customer, subscription, transaction, usage, support, feedback and churn data was modeled to create an interactive dashboard.

Key DAX measures were created for Total Customers, Active Customers, Churned Customers, Churn Rate, Retention Rate, Average Tenure and Monthly Revenue at Risk.

The dashboard enables analysis of churn across customer segments, regions, subscription plans, contract types, tenure, service usage and customer experience, while also highlighting the financial impact of customer churn.

Interactive slicers allow users to filter the dashboard and dynamically evaluate customer retention performance across different business dimensions.

MySQL Analysis & Dataset Limitation

NOTE :

Due to local MySQL Workbench import limitations, a smaller representative sample of the project dataset was used for SQL query execution and validation. The database schema and SQL logic were designed to be applicable to the larger project dataset. This approach allowed the business questions and analytical logic to be developed and tested while the complete dataset was used for the broader Excel and Power BI analysis.

SQL Analysis Performed on Business Questions :

1. Customer and segment analysis
2. Overall churn-rate calculation
3. Churn by customer segment
4. Revenue analysis by subscription plan
5. Payment-method analysis
6. Service-usage and churn analysis
7. Churn-reason analysis
8. Customer satisfaction analysis
9. Top customers by revenue
10. Subscription-plan revenue ranking
11. Revenue at risk from churn
12. Regional revenue-risk analysis

Key Insights :

- 10000 total customers are present in the analyzed customer base, providing a broad view of customer retention performance.
- 28.75% overall churn rate indicates a significant customer-retention challenge.
- 2875 customers have churned, creating a substantial impact on the customer base.
- Churn varies across customer segments, helping identify groups that require greater retention attention.
- Service usage level shows different churn patterns, making customer engagement an important retention factor.
- Churn reasons reveal the major areas contributing to customer loss and provide direction for improvement.
- Approximately ₹4.62M in monthly revenue is at risk, showing the financial impact of customer churn.
- Regional revenue at risk differs across locations, allowing the business to prioritize regions with greater financial exposure.

Key Recommendations :

- Develop targeted retention campaigns for high-churn customer segments.
- Monitor low-engagement customers and take proactive action before they churn.
- Investigate the major churn reasons and address the underlying service or customer-experience issues.
- Prioritize retention of high-value customers who contribute significantly to recurring revenue.
- Focus improvement efforts on regions with high churn and high revenue at risk.
- Investigate failed payment transactions and introduce proactive payment support where required.
- Build a future predictive churn model using Python and machine learning to identify customers at high risk of churn.

Key Limitations :

- The project uses a sample/simulated business dataset, so the findings do not represent actual company performance.
- The complete large dataset could not be imported into the local MySQL Workbench environment because of local import limitations.
- A smaller representative dataset was used to execute and validate the SQL queries.
- SQL results from the sample dataset should not be presented as results from the full dataset.
- The analysis identifies patterns and relationships, but does not prove that a factor directly causes churn.
- External factors such as competitor pricing, market conditions and customer interactions are not included.
- The project focuses on descriptive and diagnostic analytics; predictive churn modeling is planned as a future enhancement.

Overall, the analysis provides a data-driven view of customer churn, helping the business identify high-risk customer groups, understand churn patterns and quantify the potential revenue impact. The insights can be used as a foundation for targeted customer-retention strategies and future predictive analytics.

Future Scope :

- Develop a predictive churn model using Python and machine learning to identify customers at high risk of churning.
- Build customer-level risk scores to support proactive retention campaigns.
- Perform Customer Lifetime Value (CLV) analysis to prioritize high-value customers.
- Implement automated Power BI reporting and alerts to continuously monitor churn and revenue at risk.
- Extend the analysis with cohort analysis and customer behaviour trends to understand long-term retention patterns.